

Name:Hamid Reza Ghahramani

Research Question:

In patients with epilepsy (particularly drug-resistant epilepsy, West Syndrome, and Focal Cortical Dysplasia), does administration of Bumetanide (as an NKCC1 inhibitor), compared to placebo or standard anti-epileptic drugs (e.g., levetiracetam, valproate), result in a reduction in seizure frequency, changes in the NKCC1/KCC2 expression ratio, improved EEG findings, and better treatment response?

Part1:

- Based on this research question, extract the concepts and design the PICO/PECO table.
- also provide the inclusion and exclusion criteria in a table.

Element	Concept	Details/Operationalization
P (Population)	Patients with epilepsy	Specifically: Drug-resistant epilepsy (DRE), West Syndrome (infantile spasms), and Focal Cortical Dysplasia (FCD).
I (Intervention)	Bumetanide administration	Used as an NKCC1 inhibitor (often as add-on/adjunct therapy).
C (Comparator)	Placebo or Standard of Care	Placebo, or standard anti-epileptic drugs (AEDs) such as levetiracetam or valproate.
O (Outcomes)	Clinical and Biological Efficacy	Primary: Reduction in seizure frequency. Secondary: Changes in NKCC1/KCC2 expression ratio, improved EEG findings, better treatment response.

Criteria Type	Inclusion Criteria	Exclusion Criteria
Population	- Confirmed diagnosis of epilepsy (specifically Drug-Resistant Epilepsy, West Syndrome, or Focal Cortical Dysplasia). - Patients failing standard anti-epileptic drug therapy (for DRE cohorts).	- Patients with epilepsy types outside the scope (e.g., absence epilepsy not linked to FCD/DRE). - Patients with acute symptomatic seizures (provoked).
Intervention	- Administration of Bumetanide (any route/dose) intended for CNS effect (NKCC1 inhibition).	- Use of Bumetanide solely for non-neurological indications (e.g., heart failure, edema) without epilepsy monitoring.

Criteria Type	Inclusion Criteria	Exclusion Criteria
Comparator	- Studies comparing against placebo, standard AEDs (levetiracetam, valproate), or baseline (pre-treatment).	- Studies lacking a control group or baseline comparison (unless single-arm efficacy is the specific focus).
Outcomes	- Reporting of seizure frequency, NKCC1/KCC2 ratio data, EEG findings, or treatment response metrics.	- Studies reporting only pharmacokinetic data without clinical or biological efficacy outcomes.
Study Design	- Clinical trials (RCTs, open-label), observational studies, or case series with sufficient data.	- Reviews, editorials, animal-only studies (unless specifically requested for background), or conference abstracts without full data.
Language/Date	- English language (typically). - Published within a relevant timeframe (e.g., last 10–15 years for current AED standards).	- Non-English publications (if translation unavailable).

Part2:

EndNote Stage:

- Import all 10 articles into EndNote.
- Check for missing information and remove duplicate records.

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Conti, 2011 #9

Conti, 2011 #18

Tags: Manage tags

Reference Type: Journal Article

Author: Conti, L., Palma, E., Roselli, C., Lauro, C., Cipriani, R., de Groot, M., Aronica, E., Limatola, C.

Year: 2011

Title: Anomalous levels of Cl⁻ transporters cause a decrease of GABAergic inhibition in human peritumoral epileptic cortex

Journal: Epilepsia

Added to Library: 21/06/1405 Last Updated: 21/06/1405

2011 #18 Summary Edit PDF

Anomalous levels of Cl⁻ transporters cause a decrease of GABAergic inhibition in human peritumoral epileptic cortex

Conti, L., Palma, E., Roselli, C., Lauro, C., Cipriani, R., de Groot, M., Aronica, E., Limatola, C.

1528-1167/2011.03111.x

Several factors contribute to epileptogenesis in brain tumors, including reduced γ-aminobutyric acid (GABA)ergic inhibition. In particular, changes in Cl⁻ homeostasis in peritumoral microenvironment, together with

(2011). "Anomalous levels of Cl⁻ transporters cause a decrease of GABAergic inhibition in human peritumoral epileptic cortex." *Epilepsia* 52(9): 1635-1644. Several factors contribute to epileptogenesis in patients with brain tumors, including reduced γ-aminobutyric acid (GABA)ergic inhibition. In particular, changes in Cl⁻ homeostasis in peritumoral microenvironment, together with alterations of Cl⁻ homeostasis could be involved in tumor cell migration and metastasis formation. In neurons, the regulation of intracellular Cl⁻ concentration ([Cl⁻]_i) is mediated by NKCC1 and KCC2 transporters. NKCC1 increases while KCC2 decreases [Cl⁻]_i. Experiments were thus designed to investigate whether, in human epileptic peritumoral cortex, alterations in the balance of NKCC1 and KCC2 activity may decrease the hyperpolarizing effects of GABA, thereby contributing to epileptogenesis in human brain tumors.

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9 References

Author	Year	Title	Journal	Last Updated	Reference Type
Silayeva, L.; Deeb, T. Z.; ...	2015	KCC2 activity is critical in limiting the onset and severity...	Proc Natl Acad Sci U S A	21/06/1405	Journal Article
Conti, L.; Palma, E.; Rose...	2011	Anomalous levels of Cl ⁻ transporters cause a decrease ...	Epilepsia	24/06/1405	Journal Article
Chang, M.; Dufour, S.; C...	2019	Generation and On-Demand Initiation of Acute Ictal Ac...	J Vis Exp	21/06/1405	Journal Article
Blauwblomme, T.; Dzhal...	2018	Transient ischemia facilitates neuronal chloride accumu...	Ann Clin Transl Neurol	21/06/1405	Journal Article
Blauwblomme, T.; Dossi...	2019	Gamma-aminobutyric acidergic transmission underlies L...	Ann Neurol	21/06/1405	Journal Article
Berdichevsky, Y.; Sapoz...	2016	Staged anticonvulsant screening for chronic epilepsy	Ann Clin Transl Neurol	21/06/1405	Journal Article
Ben-Ari, Y.; Delpeire, E...	2021	Phenobarbital, midazolam, bumetanide, and neonatal s...	Epilepsia	21/06/1405	Journal Article
Aronica, E.; Boer, K.; Red...	2007	Differential expression patterns of chloride transporters...	Neuroscience	21/06/1405	Journal Article
Almeida, A. C.; Scorza, F...	2011	Combined effect of bumetanide, bromide, and GABAer...	Epilepsy Behav	21/06/1405	Journal Article

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Conti, L.; Palma, E.; Rose...	2011	Anomalous levels of Cl ⁻ transporters cause a decrease ...	Epilepsia	24/06/1405	Journal Article
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Blazek-Lomme, T.; Dohal...	2018	Transient ischemia facilitates neuronal chloride accumu...	Ann Clin Transl Neurol	21/06/1405	Journal Article
Blazek-Lomme, T.; Doss...	2019	Gamma-aminobutyric acidergic transmission underlies L...	Ann Neurol	21/06/1405	Journal Article
Berdichevsky, Y.; Sapozh...	2016	Staged anticonvulsant screening for chronic epilepsy	Ann Clin Transl Neurol	21/06/1405	Journal Article
Ben-Ari, Y.; Delisle, E.	2021	Phenobarbital, midazolam, bumetanide, and neonatal ...	Epilepsia	21/06/1405	Journal Article
Aronica, E.; Boer, K. Red...	2007	Differential expression patterns of chloride transporters...	Neuroscience	21/06/1405	Journal Article
Almeida, A. C.; Scorza, F...	2011	Combined effect of bumetanide, bromide, and GABAer...	Epilepsy Behav	21/06/1405	Journal Article

2011 #2

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PMCID

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Article Number

Accession Number

21167788

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Label

Keywords

Bumetanide/therapeutic use
Bumetanide/therapeutic use
Drug Therapy, Combination
Epilepsy/drug therapy
GABA Agonists/therapeutic use
Humans
Sodium Potassium Chloride
Symporter Inhibitors/therapeutic use

Abstract

Notes

1525-5069
Almeida, Antônio-Carlos G
Scorza, Fábio A.
Rodrigues, Antônio M
Arida, Ricardo M
Carlesso, Fernanda N
Batista, Alene G
Duarte, Mario A
DeCosta, Jaderson C
Letter
Research Support, Non-U.S. Gov't
United States
2010/12/21
Epilepsy Behav. 2011 Jan;20(1):147-9.
doi: 10.1016/j.epbeh.2010.10.030.
Epub 2010 Dec 21.

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Combined effect of bumetanide, bromide, a...

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Letter to the Editor

Combined effect of bumetanide, bromide, and GABAergic agonists: An alternative treatment for intractable seizures

The epilepsies constitute one of the most common serious brain disorders and know no geographic, racial, or racial boundaries, occurring in men and women and affecting people of all ages, though more frequently affecting young people in the first two decades of life and people over the age of 60 [1,2]. Worldwide, there are at least 50 million people who have epilepsy, and many of these persons have seizures that are refractory to treatment with the currently available therapies [1,3,4]. The most common risk factors for epilepsy are cerebrovascular disease, brain tumors, alcohol, traumatic head injuries, genetic inheritance, and malformations of cortical development [5,6]. In resource-poor countries, endemic infections such as malaria and neurocysticercosis seem to be major risk factors [7]. The paradoxical excitatory action of GABA observed in the immature brain plays a role in neuronal development [8,9], but also is responsible for the hyperexcitability of the immature brain [10].

chloride in the intracellular space [15], and the hyperexcitability, to the neuroepileptic coupling promoted by the decrease in extracellular space volume [16,17]. Neurophysiological studies of the molecular mechanisms that underlie the hyperexcitability and synchronization of neurons have a profitable role in the search for targets for novel antiepileptic drugs to control seizures that affect mainly infants. The treatment of refractory seizures, normally associated with intracellular chloride accumulation, involves a complex interaction of the mechanisms responsible for controlling the intracellular chloride level. Combining multiple drugs, each one targeting a different mechanism, is a promising powerful strategy to be investigated. It can be conjectured that more efficient action would be coupled with fewer side effects, as each compound would be used at a low dose, preserving their synergistic effect on the system. Initially, attention to carbon-chloride co-transporters (CCCs) focused on their physiological roles in the recovery of cell volume after swelling or shrinkage in hypotonic or hypertonic media [18]. Later, attention turned to the importance of CCCs in regulation of intracellular Cl⁻ in the central nervous system (CNS) [19]. In mature brain, the normal level of intracellular chloride must be low enough to establish inhibitory GABAergic neurotransmission. This condition is due to neuronal NaCC1

2011 #2

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Reference Type

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Author

Almeida, A. C.
Scorza, F. A.
Rodrigues, A. M.
Arida, R. M.
Carlesso, F. N.
Batista, A. G.
Duarte, M. A.
DeCosta, J. C.

Year

2011

Title

Combined effect of bumetanide, bromide, and GABAergic agonists: an alternative treatment for intractable seizures

Journal

Epilepsy Behav

Volume

20

Part/Supplement

Issue

1

Pages

147-9

Start Page

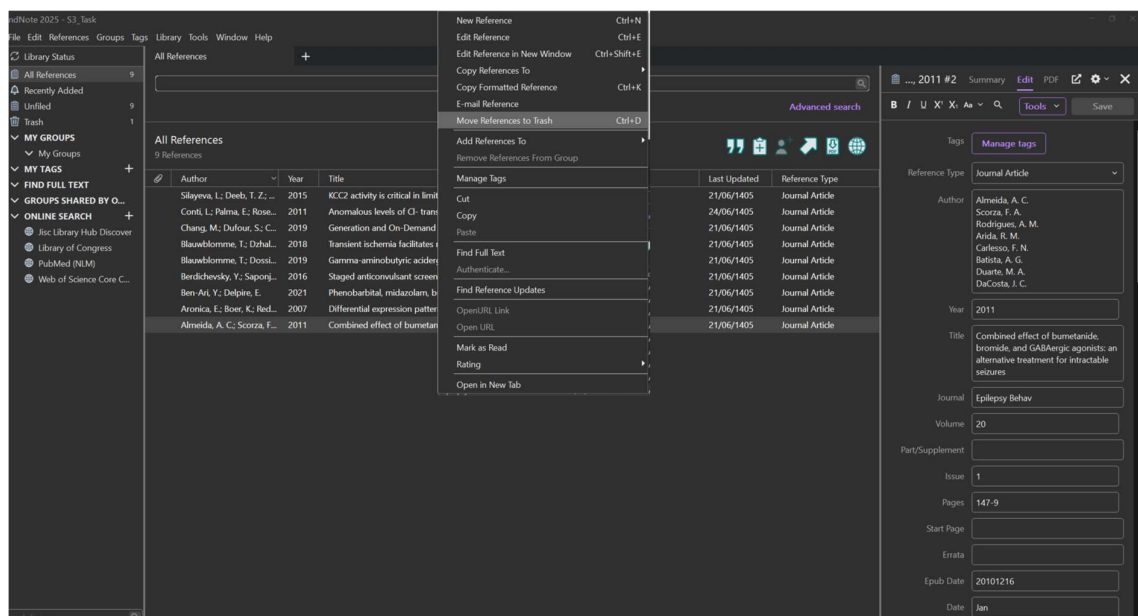
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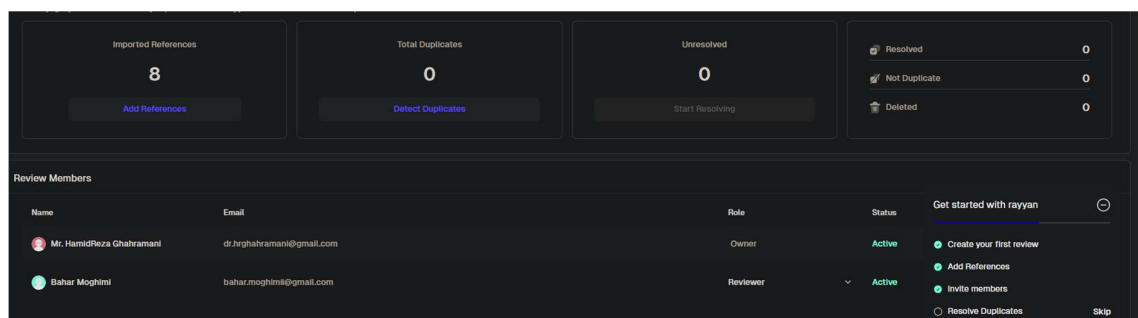
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Rayyan Stage:

- Export the cleaned records from EndNote.
- Import the file into Rayyan.
- Begin Title/Abstract screening.
- Proceed to Full-text screening.
- Complete the PRISMA table based on final results.



Rayyan | Full Text Screening

In patients with epilepsy (particularly drug-resistant epilepsy, West Syndrome, and Focal Cortical Dysplasia), does administration of Bumetanide (as an NKCC1 inhibitor)...

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01	Anomalous levels of Cl ⁻ transporters cause a decrease of GABAergic inhibition in human perih...	2011-09-01	Conti, L., Palma, E., Ro...	Attached	☒ HamidReza
02	Transient ischemia facilitates neuronal chloride accumulation and severity of seizures	2018-09-01	Blauwblomme, T., Ozh...	Attached	☒ HamidReza
03	Gamma-aminobutyric acidergic transmission underlies interictal epileptogenicity in pediatric fo...	2019-02-01	Blauwblomme, T., Des...	Attached	☒ HamidReza
04	Phenobarbital, midazolam, bumetanide, and neonatal seizures: The devil is in the details	2021-04-01	Ben-Ari, Y., Delpech, E.	Attached	☒ HamidReza
05	Differential expression patterns of chloride transporters, Na ⁺ -K ⁺ -2Cl ⁻ -cotransporter and K ⁺ -Cl ⁻ ...	2007-03-02	Aronica, E., Boer, K., R...	Attached	☒ HamidReza

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